

1. IT Infrastructure

Definition

IT infrastructure is the collection of hardware, software, networks, data storage, facilities, and IT services that an organisation uses to operate its information systems. It forms the foundation that allows business applications, communication systems, databases, and online services to function effectively.

Main Components

- Hardware (servers, computers, routers, switches)
- Software (operating systems, applications, databases)
- Networks (LAN, WAN, Internet connectivity)
- Data storage (cloud storage, databases, backup systems)
- Data centres and cloud platforms
- Security systems (firewalls, antivirus, encryption)
- IT support services and personnel

Importance of IT Infrastructure

- Supports daily business operations
 - Enables communication and collaboration
 - Stores and protects business data
 - Supports scalability as businesses grow
 - Improves productivity and reliability
 - Enables cloud computing and AI technologies
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2. Infrastructure Management

Definition

Infrastructure management is the process of monitoring, maintaining, securing, and optimising an organisation's IT infrastructure to ensure systems remain reliable, efficient, and available. It involves managing hardware, software, networks, storage, security, and cloud resources.

Key Responsibilities

- Monitoring system performance

- Maintaining servers and networks
- Applying software updates and patches
- Managing backups and disaster recovery
- Protecting systems against cyber threats
- Managing cloud infrastructure
- Capacity planning for future growth
- Troubleshooting technical issues

Benefits

- Reduces system downtime
 - Improves performance
 - Enhances security
 - Ensures business continuity
 - Supports business growth
 - Reduces operational costs through efficient resource management
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Infrastructure Management Issues Related to Meta and the Threads App

When **Meta** launched **Threads**, the application gained over **100 million users within its first five days**, creating major infrastructure management challenges.

1. Scalability

Threads experienced extremely rapid user growth. Meta's infrastructure had to automatically increase computing power, storage, and network capacity to accommodate millions of new users without major service interruptions.

Challenge: Preventing server overload while maintaining fast response times.

2. High Availability

Since Threads is a global platform, users expect access 24 hours a day.

Meta needed:

- Redundant servers
- Load balancing

- Failover systems
- Multiple data centres

These measures help ensure the platform remains available even if one server or region experiences problems.

3. Performance Management

Millions of users simultaneously:

- Posted text updates
- Uploaded images
- Viewed feeds
- Sent replies

Infrastructure managers had to continuously monitor:

- Server performance
- Database response times
- Network traffic
- Application speed

Poor performance could cause slow loading or application crashes.

4. Security

Threads stores personal information linked to Instagram accounts.

Infrastructure management includes:

- Protecting user accounts
- Preventing hacking attempts
- Encrypting user data
- Monitoring suspicious activity
- Managing authentication securely

Security becomes increasingly important as the user base grows.

5. Integration with Instagram

Threads was designed to work closely with Instagram.

Infrastructure management had to ensure:

- Fast account sign-in
- Synchronised user profiles
- Shared authentication systems
- Reliable communication between the two platforms

Any failure in integration could affect both services.

6. Data Storage

Millions of daily posts, likes, comments, and media uploads require enormous storage capacity.

Infrastructure managers must:

- Expand storage as demand increases
 - Perform regular backups
 - Ensure quick data retrieval
 - Protect against data loss
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7. Disaster Recovery

Infrastructure failures can occur because of hardware faults, cyberattacks, or power outages.

Meta must maintain:

- Backup systems
 - Data replication across multiple locations
 - Recovery procedures that restore services quickly if failures occur
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8. Cost Management

Supporting hundreds of millions of users requires substantial investment in:

- Servers

- Networking equipment
- Cloud services
- Electricity
- Data centres
- Technical staff

Infrastructure managers aim to balance high performance with cost efficiency.

Summary

IT Infrastructure

The technology resources (hardware, software, networks, storage, cloud services) that support an organisation.

Provides the foundation for business operations.

Infrastructure Management

The processes used to maintain, monitor, secure, and optimise those technology resources.

Ensures the infrastructure remains reliable, secure, and efficient.